AEVAR OEFJOERD

Plainfield, IL • (812) 204-6919 • ofjord99@gmail.com • linkedin.com/in/aevaroefjoerd • github.com/AevarOfjord

U.S. Permanent Resident (Green Card) – no sponsorship needed

SUMMARY

Mechanical Engineer with an M.S. in Mechanical Engineering and hands-on experience designing, building, and validating complex mechanical and electromechanical systems from concept through implementation. Background spans mechanical design, controls and robotics, fluid and thermal systems, simulation, automation, and industrial project engineering. Developed and validated a NASA-sponsored MPC-controlled satellite testbed, achieving 0.024 m position error, 0.51° attitude error, and less than 2% simulation-to-hardware deviation. Experienced leading multidisciplinary projects, solving complex engineering problems, and delivering systems in active industrial environments.

EDUCATION

University of Kentucky — Lexington, KY
M.Sc. Mechanical Engineering | GPA 3.63 — Aug 2024 - Dec 2025

University of Southern Indiana — Evansville, IN
B.Sc. Mechanical Engineering — Aug 2020 - May 2024

EXPERIENCE

Blue Lagoon Iceland — Grindavík, Iceland
Project Engineer/Project Manager (Contractor) — Apr 2026 - Present

- Took over a multi-million-dollar renovation and construction program mid-flight when the project lead fell ill; held schedule and coordinated work across multiple trades and stakeholders while preserving operations and guest experience.
- Serving as interim Head of Maintenance and Facilities after the department head departed; own maintenance operations, facilities projects, and contractor coordination; cut the pending work-order backlog from 200 to 125 (38% reduction) while supporting the transition to newly appointed leadership.
- Designed 1 km of wastewater piping for the geothermal spa (routing, backup routes, pump selection) plus a 550 m cold-water supply network with flow simulation, valve sizing, and pump specifications to ensure reliable delivery and controllability.
- Redesigned the intake and heating system for a large recirculating lagoon: isolated supply from the adjacent geothermal plant for independent flow control and rerouted through high-temperature flash heaters, improving thermal performance and operational flexibility.
- Led Computational Fluid Dynamics (CFD) analysis for an enclosed recirculating cave pool to resolve poor water renewal; evaluated multiple inlet configurations and coordinated implementation of a fresh supply solution to improve water quality.
- Selected and rolled out Connecteam for construction management and took ownership of MaintainX, configuring workflows, asset hierarchies, and safety tracking to support ISO 9001 and ISO 45001 compliance across facilities.
- Designing an AI project manager for maintenance operations that integrates MaintainX with a Microsoft Copilot Studio workflow of specialized agents: triages incoming work requests, converts them into scheduled work orders, monitors active jobs, escalates delays, and routes exceptions to a human project manager.

University of Kentucky — Lexington, KY
Graduate Research Assistant — Jan 2025 - Aug 2025

- Designed and built a planar satellite testbed on an air-bearing platform with 8 binary cold-gas thrusters (2000 psi tanks regulated to 40 psi); served as the primary validation environment for real-time constrained MPC research.
- Developed and implemented a real-time mixed-integer MPC controller (mixed-integer quadratic programming (MIQP) formulation, Gurobi) operating at 16.7 Hz under binary actuator and computational constraints, achieving a final position error of 0.024 m and final attitude error of 0.51° on physical hardware.
- Built a high-fidelity digital twin with shared controller code and matched physical parameters (including individually characterized thruster forces with ~13% variation); achieved <2% deviation in total thruster usage between simulation and hardware over full multi-minute missions.
- Built Python simulation, visualization, logging, and hardware-in-the-loop (HIL) tooling with OptiTrack state estimation; demonstrated solver feasibility (avg 4.61 ms, worst-case 32.1 ms) with no missed control updates at 60 ms intervals.

Blue Lagoon Iceland — Grindavík, Iceland
Maintenance Software Specialist (Contractor) — Jul 2025 - Aug 2025

- Led full migration from legacy computerized maintenance management system (CMMS) to MaintainX across 20+ facilities, designing asset hierarchies, QR-coded asset tagging, preventive maintenance workflows, user permissions, and team structures aligned with operational requirements.
- Standardized and digitized maintenance processes facility-wide by configuring checklists, work orders, and documentation libraries in MaintainX, improving scheduling efficiency, data consistency, and operational visibility.

Blue Lagoon Iceland — Grindavík, Iceland
Project Manager — May 2022 - Jan 2025

- Managed facility maintenance programs across geothermal spa operations, coordinating vendors, contractors, and internal teams.
- Delivered facility maintenance projects on time under severe operational constraints, including sustained execution during the 2023 Grindavík volcanic crisis.
- Led planning meetings, defined project requirements, and tracked execution across concurrent workstreams.

University of Kentucky Lexington, KY

Teaching Assistant – Machine Design Aug 2024 - Dec 2024

- Provided technical feedback on stress analysis, failure criteria, fatigue, fracture, and component design problems for undergraduate Machine Design coursework.

Blue Lagoon Iceland Grindavík, Iceland

Maintenance Technician May 2017 - May 2022

- Performed preventive and corrective maintenance on mechanical, pneumatic, and fluid systems across geothermal spa facilities.
- Diagnosed and repaired equipment failures, maintaining service records across 20+ facility assets.

PROJECTS

3D MPC-Controlled Inspection Satellite Jan 2026 - Mar 2026

github.com/AevarOfjord/3D-MPC-Inspection-Satellite

- Extended planar testbed architecture into full six-degree-of-freedom (6-DOF) motion with coupled position and attitude control for inspection trajectories using MPC. Integrated thruster-based translation with reaction wheel-based attitude control in a unified architecture.
- Built browser-based Mission Studio for trajectory design and mission configuration; developed simulation, visualization, and telemetry tools with automated sweeps for control interval and horizon tuning.

2D MPC-Controlled Inspection Satellite (M.Sc. Thesis) Jan 2025 - Nov 2025

github.com/AevarOfjord/mpc-air-bearing-satellite | doi.org/10.13023/etd.2025.583

- Designed and built the complete platform end to end: mechanical structure, air-bearing system, 8 binary cold-gas thrusters, ESP32 embedded control, and Python/C++ software stack. Published as M.Sc. thesis with full controller design, digital-twin, and hardware validation results.

Nonlinear MPC for Multi-Link Inverted Pendulum Mar 2026

github.com/AevarOfjord/Inverted-Pendulum

- Developed a nonlinear model predictive control (NMPC) framework (acados/CasADi, HPIPM backend) to swing up and stabilize 1–8 linked inverted pendulums on a cart using a unified controller architecture. Achieved real-time feasibility for 1–5 link systems (avg solve <5 ms); dual Python/C++ runtime; CMA-ES weight tuning and SQP_RTI vs full SQP trade-off evaluation.

Dual-Arm Chess Robot – Control & Motion Planning System Jul 2026 - Present

github.com/AevarOfjord/Automatic-Chess

- Building a dual-arm chess robot: two opposing planar 3R arms with a PC-side controller that owns game state, move planning, inverse kinematics, and collision-aware transfer trajectories, while ESP32 firmware executes motion over a JSON command protocol. Developed a digital-twin simulator with live planned-path overlays, a geometry optimizer for link lengths and joint windows with reachability certification, and deterministic capture and fault-recovery handling; hardware build in progress.

Undergraduate Capstone – Lava Rock Processing System Jan 2024 - Apr 2024

doi.org/10.58090/usi.1491793

- Designed machinery for processing volcanic lava rock into commercial materials; performed structural and force analysis in SolidWorks and validated design through physical testing of a scaled functional prototype using 3D printing and real material samples.

SKILLS

Control & Optimization: Model Predictive Control (MPC), Nonlinear MPC (NMPC), Mixed-Integer Linear/Quadratic Programming (MILP/MIQP), Sequential Quadratic Programming (SQP/SQP_RTI), Gurobi, acados, CasADi, HPIPM, CMA-ES

Robotics & Autonomy: State Estimation, Trajectory Planning, Hardware-in-the-Loop (HIL), Real-Time Systems, OptiTrack, Digital Twins

Programming & Embedded: Python, C++, MATLAB, Git/GitHub, ESP32, Arduino, Raspberry Pi

Design & Simulation: SolidWorks, Fusion 360, 3D Printing, Finite Element Analysis (FEA, SolidWorks Simulation), Computational Fluid Dynamics (CFD, OpenFOAM), Piping & Thermal Systems

Project Systems & Tools: MaintainX, Connecteam, Contractor Coordination, Microsoft Office (Word, Excel, PowerPoint, Power Automate, Planner, Teams, Microsoft Copilot Studio)

LEADERSHIP & ACTIVITIES

NCAA Division I Varsity Soccer – University of Southern Indiana (2020–2024) • Senior Team Player – Grindavík and Víðir Soccer Clubs, Iceland (2017–2023) • Academic Honor Roll (multiple terms); Academic All-GLVC and OVC Commissioner's Honor Roll.